Ogbonna Arinze

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SUMMARY

Frontend-focused fullstack engineer with 8 years of experience building high-performance, data-intensive web applications across healthcare, fintech, and civic tech. Deep expertise in React, TypeScript, and Next.js, with strong backend capabilities in Node.js, PostgreSQL, and Docker. Proven track record of building real-time data visualization dashboards, optimizing application performance at scale, and integrating observability tooling. Passionate about developer tools and open-source ecosystems, with hands-on experience building RAG-powered platforms and production monitoring instrumentation.

CORE COMPETENCIES

- **Frontend & UI:** React, Next.js, TypeScript, Vue.js, Redux, React Query, Zustand, Tailwind CSS, Radix UI, Recharts, react-chartjs, and responsive design systems.
- **Backend & APIs:** Node.js, NestJS, Express, REST APIs, GraphQL, JWT authentication, RBAC, SSE streaming, and WebSockets.
- **Databases & SQL:** PostgreSQL (pgvector), MongoDB, Prisma ORM, SQL query generation, and schema design.
- **Infrastructure & DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS, Traefik, Nx monorepos, and pnpm workspaces.
- **Observability & Performance:** Langfuse, Sentry, Datadog RUM, Lighthouse, Chrome DevTools, React Profiler, bundle analysis (webpack-bundle-analyzer, rollup-plugin-visualizer), and Core Web Vitals.
- **Data Visualization:** Recharts (22 chart types), react-chartjs, dynamic dashboards, and real-time data streaming with EventSource API and SSE.
- **Testing & Quality:** Playwright, Jest, Enzyme, Allure Reports, Storybook, ESLint, and CI-driven QA pipelines.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), and agentic workflows.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations with upstream EHR systems.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG compliance), and developer velocity across multiple product teams.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to query their data in real time via a conversational interface.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, measuring improvements via **Lighthouse** and **Chrome DevTools** profiling.
- Introduced **Redux** for state management and **React Query** for server-state caching, refactoring legacy class components to functional components and streamlining data fetching patterns across the application.
- Implemented **JWT-based authentication** and **role-based access control** with HTTP-only cookies for secure token handling, collaborating with backend engineers to align on auth strategies and permission-based UI rendering.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **GitHub Actions CI/CD pipeline**, publishing automated reports with **Allure Reports** on GitHub Pages and configuring Slack notifications for build visibility.
- Mentored a team of two junior developers (interns), conducting code reviews and fostering a culture of technical excellence.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to build a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse Audits**, reducing bundle size by **30-40%** through **rollup-plugin-visualizer** and **webpack-bundle-analyzer**, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms, significantly boosting speed and user experience across devices.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices with clear communication of component APIs.

- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code and consistent design patterns across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, processing thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily transaction reconciliation by **60%** through innovative data processing and management strategies.
- Enhanced system responsiveness by adopting the native browser **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster updates and lower latency.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **Node.js RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance on Node.js and React development workflows.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **Node.js**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the frontend to **Node.js** backend services, and devised comprehensive documentation detailing all application functions, capabilities, and features.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, orchestrated with **pnpm** workspaces.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring **22 interactive chart types** auto-generated from AI responses, dark mode, URL state management, and responsive mobile layouts.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including **pgvector** extension for vector storage, and wrote custom migration tooling for schema drift detection.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management, and conversation summarization to prevent reverse proxy timeouts.
- Configured **CI/CD** with **GitHub Actions**, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and **Traefik** reverse proxy routing.
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management across environments, enabling production monitoring and debugging of AI pipeline performance.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*